

Where Models Miss: A Cross-Platform Analysis of GUI Action Grounding

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Abstract

1. Introduction

2. Related Work

3. Method

Click2Act is an open, extendable evaluation framework for GUI agents. It supports two evaluation modes—point grounding and action prediction—and is designed to be modular: new agents and benchmarks are registered independently and paired at runtime via YAML configuration files as illustrated in Figure 1.

3.1. Evaluated Models

We evaluate four small pure-vision VLMs, all used with no additional fine-tuning. Table 1 summarizes each model.

Table 1. Models evaluated in this study.

Model	Params
UI-AGILE-3B	3B
Holo2-4B	4B
AGUVIS-7B	7B
AGUVIS+UI-Zoomer	7B

All evaluated models fall in the 3B–7B parameter range, targeting practical deployment on consumer devices like laptops and workstations without cloud infrastructure. Despite this appeal, how much accuracy they sacrifice compared to flagships such as UI-Venus-1.5-30B-A3B (Venus-Team et al., 2026) and Holo2-235B-A22B (Company, 2025), and where exactly they fail, remains an open question.

AGUVIS-7B (Xu et al., 2024) is a fully open-source autonomous GUI agent built on Qwen2-VL that supports all

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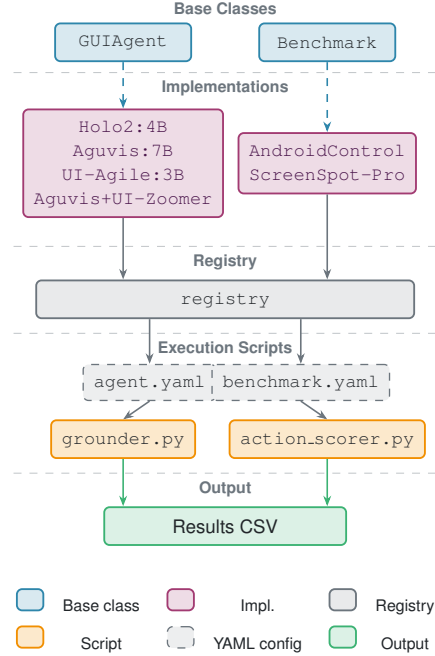


Figure 1. Click2Act framework overview. Abstract base classes (GUIAgent, Benchmark) are subclassed by concrete implementations and resolved at runtime via a registry. Each execution script receives an agent.yaml and a benchmark.yaml, runs inference over benchmark samples, and writes results to a CSV for offline analysis. Dashed arrows = inheritance; solid arrows = data flow.

platforms. It operates in two stages: (1) a planning stage where the model generates a structured *inner monologue*—consisting of a textual description of the current screen, internal reasoning toward the high-level goal, and low-level action instructions—and (2) a grounding stage where the inner monologue is translated into executable actions. It was one of the first small fully open-source vision-based GUI agents to demonstrate competitive performance on ScreenSpot.

Holo2-4B (Company, 2025) is a fully open-source autonomous GUI agent built on Qwen3-VL that supports all platforms. It is trained in two stages: (1) a large-scale supervised fine-tuning (SFT) stage on screen localization and element grounding, and (2) an online reinforcement learning

stage using Group Relative Policy Optimization (GRPO) to refine its multi-step reasoning and navigation policies.

UI-Agile-3B (Lian et al., 2025) is a fully open-source GUI grounding agent built on Qwen2.5-VL-3B. It is aiming at improving both the reinforcement fine tuning and the inference stages of GUI agents. their contibution are in three main points: firstly, UI-Agile applies simple thinking strategy that encourages thoughts within an "ideal range" of length, secondly, UI-Agile provides a continuous reward function that treats a click on the very edge of a button the same as a click in the center, thirdly, UI-Agile applies a cropping-based resampling strategy that dynamically adjusts the difficulty by cropping (zooming in) on the original screenshot.

AGUVIS + UI-Zoomer we augments AGUVIS-7B with UI-Zoomer (Tang et al., 2026), an inference-time post-processing technique that requires no retraining. When the model’s localization confidence is low, UI-Zoomer crops and zooms into predicted region, adding an extra forward pass only when needed. We pair UI-Zoomer to test whether inference-time augmentation can close the accuracy gap without increasing model size.

3.2. Expirmented Benchmarks

ScreenSpot-Pro (Li et al., 2025) is a static GUI grounding benchmark targeting high-resolution desktop screenshots, with 1,580 samples distributed across 26 professional applications on Linux, macOS, and Windows. State-of-the-art models have not yet saturated its leaderboard, making it a strong discriminator between models. Each sample pairs a screenshot with a low-level instruction and a ground-truth bounding box. Figure 2 shows the distribution of samples across applications and UI element types. Text targets out-number icon targets in most applications, with the gap being especially large in domain-specific tools such as `matlab`, `solidworks` and `word`.

Android Control (Li et al., 2024) is a mobile GUI dataset. We evaluate on a subset of episodes from its test set, where each episode consists of a high-level goal and multiple steps. Each step pairs a screenshot and a low-level instruction with a ground-truth action (action type, coordinate, text, and scroll direction).

3.3. Evaluation Metrics

Hit Rate (ScreenSpot-Pro) measures the fraction of samples for which the predicted point falls within the ground-truth bounding box.

Step Accuracy (Android Control) measures the fraction of steps for which the predicted action matches the ground truth on all applicable components. Coordinates are normalized to $[0, 1]$ by the screen dimensions, so a predicted coordinate

(x', y') is considered correct if $\|(x', y') - (x^*, y^*)\|_2 \leq \tau$, where (x^*, y^*) is the ground-truth coordinate. We report results at $\tau \in \{0, 0.01, 0.05, 0.1\}$ to characterize each model’s coordinate precision across a range of tolerance levels.

Component-level accuracy decomposes step accuracy into component scores (action type, coordinate, text, direction), enabling identification of which prediction components are responsible for failures.

4. Experiments

5. Conclusion

Acknowledgements

Impact Statement

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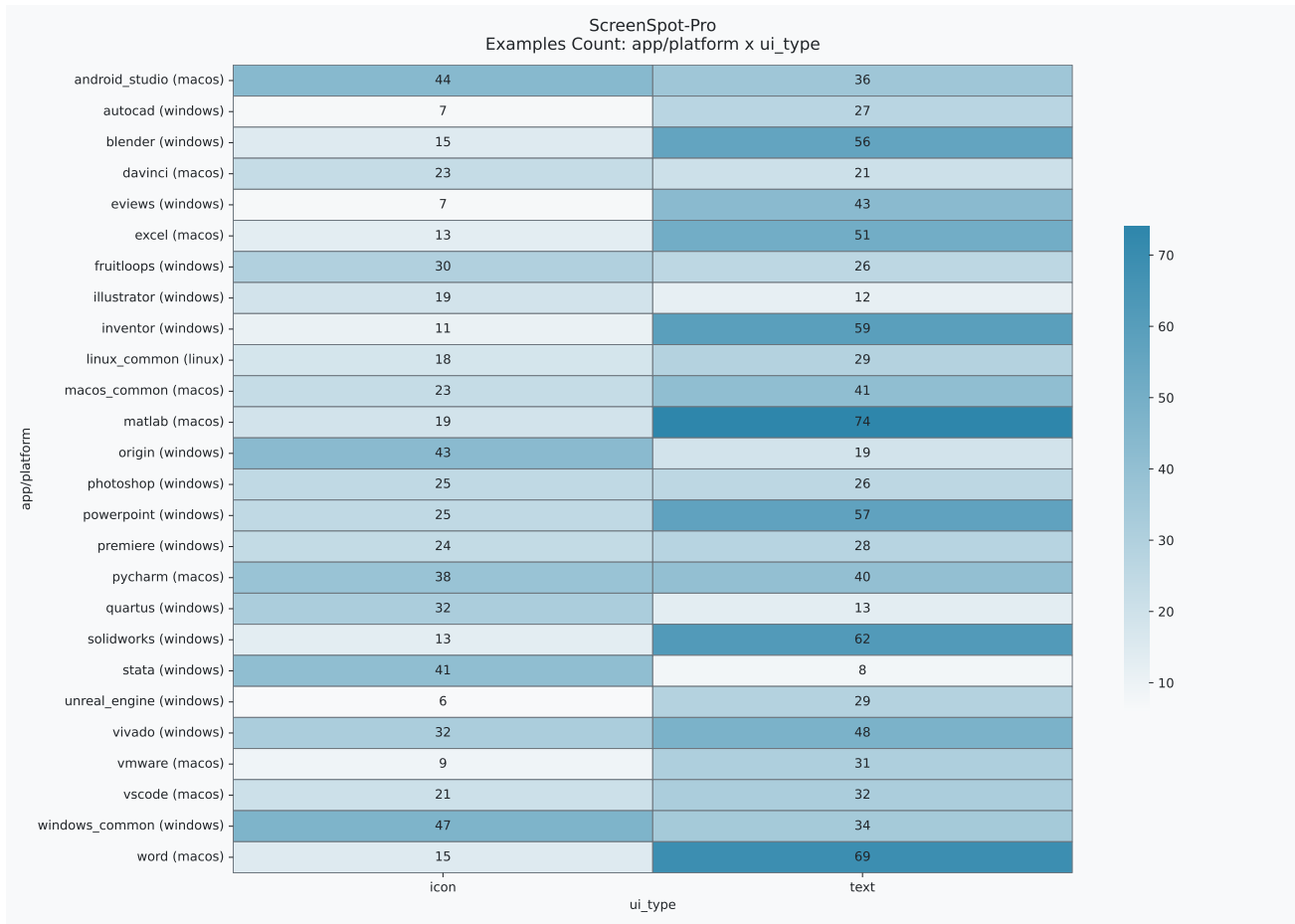


Figure 2. ScreenSpot-Pro sample distribution across applications and UI element types. Text targets outnumber icon targets in most applications.

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(Xu et al., 2024) (?)

A. Appendix